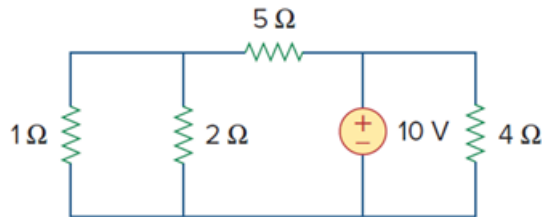


Problem

How many branches and nodes does the circuit in Fig. 2.14 have? Identify the elements that are in series and in parallel.

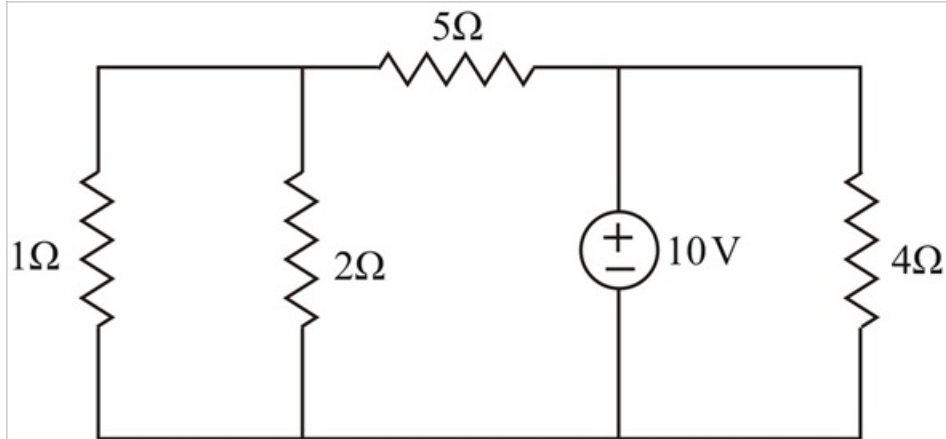
Figure 2.14:



Step-by-step solution

Step 1 of 3

The given circuit is,

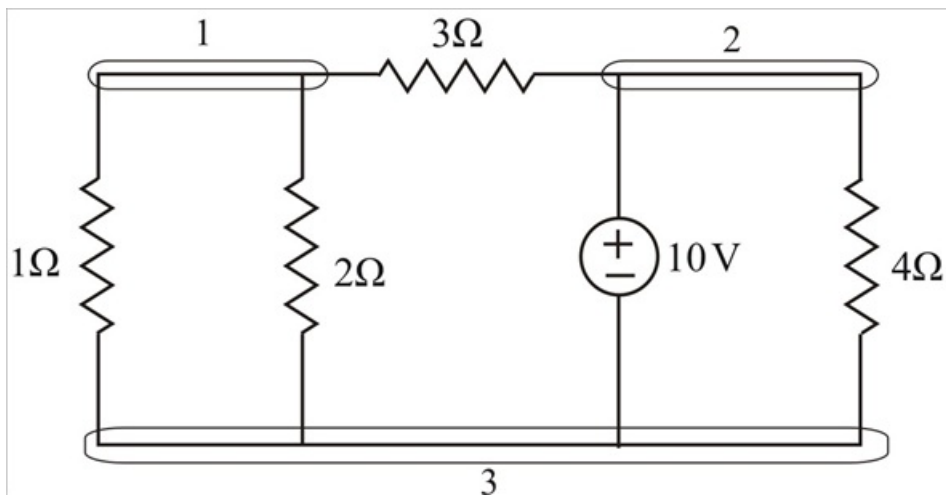


Step 2 of 3

Since there are five elements in the circuit, the circuit has **five branches**:

$1\ \Omega$, $2\ \Omega$, $5\ \Omega$, $4\ \Omega$ and $10\ \text{V}$.

The circuit has **three nodes** as identified in below figure,



Step 3 of 3

The 1Ω and 2Ω resistors are in parallel because both are connected to the same nodes 1 and 3 .

The 4Ω resistor is in parallel with the $10V$ voltage source because both are connected to the same nodes 2 and 3 .